

SAMAIL, Oman

WMO: 412570

Lat: 23.254N

Lon: 57.928E

Elev: 1368

StdP: 13.98

Time Zone: 4.00 (E04)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	55.2	57.6	23.3	18.5	89.8	27.6	22.6	87.5	19.0	72.8	16.1	70.8	1.7	210	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	18.6	112.2	70.3	110.5	70.9	108.8	71.2	81.5	97.7	80.2	96.9	79.0	96.3	10.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	150.2	91.6	75.8	142.1	91.5	74.0	133.5	91.5	46.6	97.4	45.1	97.3	43.8	96.1	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.1	13.4	DB	49.6	115.8	2.3	1.2	47.9	116.7	46.5	117.3	45.2	118.0	43.6	118.8	(4)
(5)			WB	42.1	83.2	2.9	1.7	40.0	84.5	38.3	85.5	36.7	86.4	34.5	87.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.5	68.1	72.3	79.0	87.0	95.4	99.2	99.1	97.1	93.5	86.8	77.4	70.1
	DBStd	11.86	3.99	5.05	5.09	4.92	4.03	3.48	3.60	3.55	2.91	3.39	3.88	3.71
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	26	15	4	0	0	0	0	0	0	0	0	0	7
	CDD50	12952	563	624	900	1111	1406	1475	1521	1461	1304	1142	823	622
	CDD65	7502	113	208	435	661	941	1025	1056	996	854	677	373	163
	CDH74	111044	560	1519	4531	9214	15621	17795	18163	16637	13512	9301	3308	883
	CDH80	74446	84	471	2083	5430	11197	13481	13702	12174	9206	5296	1184	138
Wind	WSAvg	5.6	4.4	5.3	5.5	6.0	6.8	6.9	6.7	6.1	5.8	5.0	4.4	4.0
Precipitation	PrecAvg	6.7	0.4	0.7	1.1	0.8	0.4	0.4	0.8	0.6	0.4	0.4	0.4	0.4
	PrecMax	23.3	2.8	3.7	12.1	2.9	2.3	2.5	4.2	2.6	2.6	2.0	2.6	2.2
	PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.8	0.7	0.9	2.3	0.9	0.5	0.7	1.0	0.6	0.6	0.5	0.7	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.2	92.6	98.1	104.4	112.7	114.7	114.1	112.4	107.9	103.1	93.5	86.3
		MCWB	60.2	62.8	64.4	66.2	68.1	68.2	71.1	71.7	70.0	66.3	64.7	62.7
	2%	DB	82.1	88.6	95.6	102.3	109.5	112.4	111.8	110.1	106.0	100.9	90.8	83.4
		MCWB	61.6	63.2	63.8	65.9	69.0	69.2	71.7	72.2	70.6	66.5	64.9	63.1
	5%	DB	79.4	85.4	93.0	100.0	107.1	110.6	110.0	108.0	104.3	98.7	88.7	81.2
		MCWB	61.6	62.8	63.6	66.1	68.7	69.7	72.5	72.4	70.3	66.5	65.1	63.2
	10%	DB	76.9	82.5	90.2	97.7	105.1	108.7	108.1	106.1	102.4	96.4	86.5	79.0
		MCWB	61.1	62.3	63.3	65.5	68.1	70.2	73.0	73.0	70.4	66.1	65.0	63.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.1	69.9	72.3	75.2	78.4	81.7	83.6	82.6	80.2	76.7	72.6	69.3
		MCDB	77.0	80.9	83.0	87.7	91.8	98.7	99.8	97.8	93.8	88.2	83.6	76.6
	2%	WB	65.6	67.5	70.2	73.0	76.5	79.8	81.7	81.3	78.4	74.2	70.9	67.9
		MCDB	74.7	78.6	81.5	85.8	91.8	96.9	98.6	96.6	93.4	88.2	81.4	75.8
	5%	WB	64.3	66.2	68.8	71.5	75.1	78.2	80.3	80.1	76.9	72.6	69.4	66.6
		MCDB	73.3	77.2	81.6	85.5	92.6	96.2	98.5	95.9	93.1	87.5	80.5	74.9
	10%	WB	63.2	64.9	67.2	70.0	73.6	76.8	78.9	78.9	75.3	71.1	68.3	65.5
		MCDB	72.6	76.9	81.7	87.5	93.5	96.2	97.8	95.7	93.8	87.5	79.8	74.2
Mean Daily Temperature Range	5% DB	MDBR	17.6	18.8	19.7	18.7	18.8	18.6	17.4	17.7	18.5	19.1	17.4	17.3
		MCDBR	20.2	21.7	22.2	20.3	20.1	20.2	19.3	18.8	19.3	20.7	18.8	20.0
		MCWBR	7.4	8.2	8.5	7.4	8.3	7.7	7.2	7.0	7.7	8.2	8.3	7.7
	5% WB	MCDBR	17.4	17.6	17.5	17.3	18.5	17.3	15.8	16.3	17.4	17.4	16.0	16.0
		MCWBR	7.5	7.5	8.4	8.1	9.6	9.3	7.5	6.9	7.4	8.3	7.9	6.7
Clear-Sky Solar Irradiance	taub	0.396	0.439	0.502	0.560	0.590	0.660	0.736	0.678	0.569	0.462	0.420	0.396	
	taud	2.261	2.104	1.916	1.810	1.748	1.637	1.543	1.632	1.813	2.049	2.197	2.251	
	Ebn at Noon	272	267	256	243	234	217	201	213	234	256	262	267	
	Edh at Noon	39	49	62	69	74	82	90	82	67	51	41	38	
All-Sky Solar Radiation	RadAvg	1459	1712	1984	2195	2388	2336	2216	2129	2015	1832	1525	1380	
	RadStd	67	76	106	116	53	83	89	55	39	51	43	65	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (13 neighbors)	+0.76	+1.22	N/A	N/A	+0.50	+0.54	N/A	N/A	+173	+176

Nomenclature: See separate page